

Customer Sentiment & Intent Classifier: Cloud & SaaS Infrastructure Domain

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Abstract— This technical report evaluates the mechanics of supervised neural text classification compared to rule-based logic. Utilizing a customer sentiment and intent classification experiment, we analyze the engineering limitations of hardcoded conditionals, evaluate the impact of training dataset diversity on model confidence, and identify the precise location of feature extraction within the standard machine learning pipeline.

Keywords—NLP, text analysis, intent classification, sentiment analysis

I. INTRODUCTION

Customers' feedback ingestion channels are highly variable and have informal grammar, domain-specific terms, and subtle semantic modifiers like sarcasm. Traditional enterprise systems have long been based on static keywords and regular expressions to perform text triage, but they do not scale with high-variance production workloads. In this activity, an experimental system that implements an Intent Classifier using a neural network is studied, which classifies incoming customer text into three operational classes: Positive Praise, Negative Complaint, Urgent Report. We examine the mechanisms in the architecture that support neural networks to learn more than words, using progressive training iterations and failure-mode analysis.

II. EXPLICIT RULES VS TAGGED LEARNING

Attempting to classify customer feedback using explicit IF/ELSE conditionals presents severe software engineering and scalability challenges:

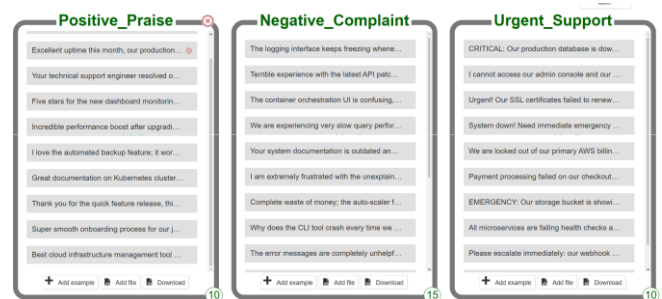
- 1.) Human language is full of almost endless syntactic variations, misspellings, and slang. Regulatory rules become unmaintainable if they are so specific as to cover all the different ways that a complaint can be formulated.
- 2.) Contextual Fragility and Keyword Collision: Hardcoded rules are based on "surface" keywords. The instruction "IF "great" IN text THEN Positive" will not work right away if the text includes sarcasm, as in "Oh great, another crash".
- 3.) Absence of Probability Scoring: Rule-based systems deliver binary responses, instead of scores, which means that the downstream systems cannot pass the ambiguous edges to human agents for manual triage.

III. DATA BIAS & DATASET QUALITY

The model's performance was found to be rather unstable, with significant fluctuations in the number of misclassifications, the level of confidence across test inputs and high variance, during early trials with only 3 examples per class. A larger set of 10 diverse examples per class was used to stabilize the performance of inferences and later on

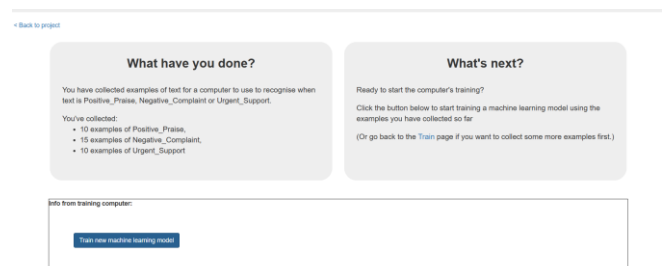
another 5 examples were added to the negative (complaint) samples.

Since neural networks are statistical function approximators, accuracy of deep learning is very dependent on the quality, quantity, and even the diversity of the training data. With small sample numbers (e.g., 3 samples), the network can over fit the vocabulary tokens and create spurious correlations. A larger number of diverse and representative examples makes the network learn to generalize across the different ways sentences are semantically structured, which allows the network to consider the context of multiple vocabulary tokens over time, rather than focusing on only a single adjective. language.



IV. CONNECTING TO THE SLIDE ARCHITECTURE: FEATURE EXTRACTION

Within the text classification pipeline, feature extraction occurs immediately between the Data Ingestion/Preprocessing phase and the Neural Network Layer (Model Training/Inference). For example for this activity, the feature extraction the moment I pressed the "train new machine learning model" button.



Automated text recognition systems do not allow raw string inputs for the dense neural network layers. The feature extraction process converts a text into a numerical vector representation in three steps:

- 1.) Tokenization: Segmentation of raw phrases into individual word or sub-word tokens.
- 2.) Normalization – remove punctuation, uniform word case, and getting the root words.
- 3.) Vectorization / Word Embedding: Encoding tokens in the form of numerical feature vectors (TF-IDF, Bag-of-Words or dense vectors).

This transformation is responsible for the association of mathematical weights with a classification label in the neural network to learn the association between phrases that can be input into the network and a classification label.

V. CONCLUSION & FUTURE WORK

The main problem of Supervised Text Classification is that it is structurally brittle, meaning that it has limited ability to recognize semantic patterns when rules are absent. The major drawback of Supervised Text Classification is its structural brittleness, which is the limited capacity to recognize a semantic pattern in the absence of rules. However, empirical results show that neural networks are still very sensitive to the sparsity of the dataset as well as to cases where the language is not covered by the dataset.

Engineering teams need to have a variety of training sets, continuously test for failures, including sarcasm, and employ probability thresholding to keep an eye on how well their inferences are going in production.